

Design and control of fast steering mirror for satellite communication process

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Abstract— Free Space Optical (FSO) communication is emerging as a method of exchanging desired information over long distances like outer space. To achieve this, the transmitting and receiving units located over a long distance must be aligned each other. There are many disturbances in free space that can interfere with the alignment. Therefore, a Fast-Steering Mirror (FSM) mechanism is applied to correct the misalignment with high precision. In this study, we designed a FSM mechanism to achieve precise beam alignment between the transmitter and receiver in optical communication systems. The mechanism has a Voice Coil Motor (VCM) as an actuator, cartwheel hinge as a rotational guide, and laser displacement sensors as tip/tilt motion sensor. The VCM was designed based on magnetostatic analysis to have sufficient actuating force in the mechanisms' motion range. The sensors was properly placed perpendicular to the mirror to measure tip/tilt motion of the mechanism. In particular, the guide was designed with a cartwheel hinge that has sufficient stiffness despite its small size to ensure wide control bandwidth. The design was performed using optimization techniques, and the result was confirmed using the finite element method. The control results indicated a wide tilt range of 1.5 °, a high resolution of 0.005 °, and a small steady state error of 1%. This approach can offer improvements in the performance of laser beam alignment. Enhancing alignment accuracy and reduced error of the FSM mechanism are also expected to be beneficial such as LIDAR and telescope.

Keywords— Fast Steering Mirror, Voice Coil Motor, Compliant Mechanism, Cartwheel Flexure, Sensor Kinematics

I. INTRODUCTION

As the era of artificial intelligence opens, the amount of data exchange is explosively increasing in long-range optical fiber data links, local/wide/metropolitan area networks (LANs/WANs/MANs), and Gbps Ethernet communication ports, and so on. To meet the demands of increased data traffic, FSO communication has been introduced to innovate networking solutions [1]. FSO is a system that transmits large diameter laser beams through the air, after electrical-to-optical conversion. It can receive the laser beam and then convert the optical signal to electrical one. While FSO systems offer advantages over wireless local area networks (LANs) in terms of security, because the laser beam does not spread like radio waves, but it is focused on intended target. Conventional FSO systems are typically point-to-point link and are subject to vulnerabilities such as distraction on the optical path vibration on the transceiver. Furthermore, the installation and maintenance of the FSO systems can be labor-intensive and time-consuming [2]. To solve the problems, various FSO systems have been studied. They have employed motor-driven

reflection mirrors for beam steering and a bundle of laser beams from optical fibers. These designs aim to maintain communication even when disturbances affect the FSO equipment by keeping the laser beam within the receiver's detection range through feedback control of the mirror and/or fibers. This method has received attention in the studies because it enables reliable and flexible long-range communication despite the distraction and the vibration. To achieve the high precision of laser beam alignment, the motor-driven reflection mirror mechanism, so called FSM is important. The mechanism comprises a mirror, actuator, and flexural hinge, which allows for precise adjustment of the laser direction by controlling tip/tilt angles of the mirror through motor rotation [3]. The actuators used in FSM can include Voice Coil Actuators (VCA) [4], Moving Magnet Voice Coil Motors (MMVCM) [5], or piezoelectric transducers (PZT) [6], each contributing to different aspects of precision and responsiveness. The flexural hinge, acting as a rotation guide, influences the system's precision and control bandwidth by enabling relative rotation between components without friction, hysteresis, or backlash [7]. Various types of flexural hinges, such as leaf, notch, cross-axis, and cartwheel, can be selected based on the desired performance characteristics [8].

II. DESIGN OF FSM

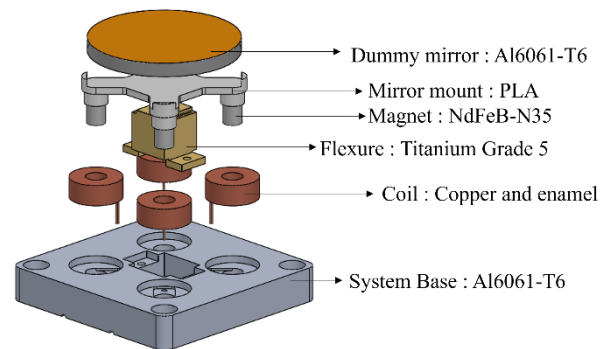


Fig. 1. Exploded view of FSM system.

Fig. 1. is an Exploded view of FSM system. It has simple shape MMVCM, cartwheel-shaped guide flexure and mirror. In this design. There must be sufficient air gaps between coils and magnets to prevent interference between the parts because the assembly of the four coil windings and mirror has a tilt motion about the pivot point of the guide.

A. Design of Actuator

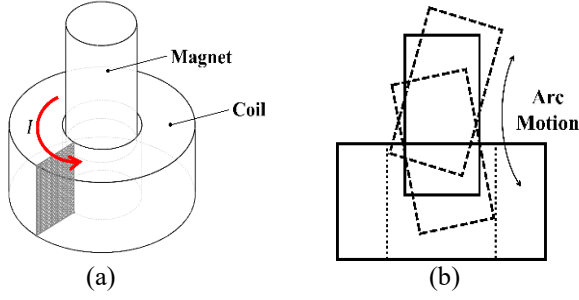


Fig. 2. Schematic of MMVCM, (a) configuration, (b) rotational motion of magnet.

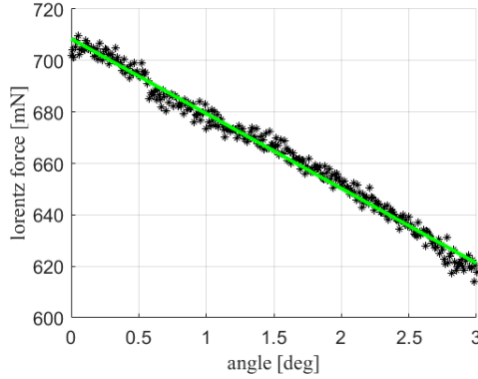


Fig. 3. MMVCM's actuating force according to the magnet's rotational motion

Fig. 2(a) shows the structure of the MMVCM. The actuator is driven by the Lorentz force between the coil and the magnet. The coil is fixed on the base and the magnet is attached to the mirror and moving. It is compactly designed to be attached to a small dummy mirror with a diameter of 64 mm. During tip/tilt motion the MMVCM rotates along the radius of the rotation center. Therefore, it is necessary to analyze the actuating force change due to the tilt angle. Fig. 3 shows the graph of the Lorentz force vs. magnet's angle. The actuating force was changed by about 12.3 % from 708.4 mN to 621.2 mN as the angle was tilted from 0° to 3°. This change can be overcome through a feedback control.

B. Flexural Hinge Design

In this section, we perform the design of a flexible hinge that acts as a rotation guide for the FSM mechanism. In order to follow the control input of the tip/tilt motion, the mechanism must be symmetric about the pivot point of the rotational guide. Therefore, we propose a cartwheel flexible hinge (CFH) with a shape as shown in Fig. 4. The corresponding hinge is a kind of pivot, which is symmetrical with respect to the tip/tilt axis direction. It can have an accurate rotation center due to cartwheel flexure shape. This flexible hinge has a relatively high stiffness with the same volume, so it has a wider control bandwidth. It also meets the requirement of the size limitation of the FSM mechanism.

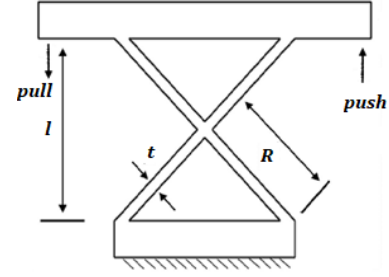


Fig. 4. 2D view of cartwheel flexure hinge

Each guiding hinge consists of two pairs of CFHs, where the first pair handles rotation around the x-axis and is attached to the base, while the second pair controls rotation around the y-axis and connects to an intermediate body. Each CFH pair linked to the base is further connected to this intermediate body, designed so that the rotation axes intersect at the origin. The base and intermediate body act as thick, rigid bodies compared to the other flexible components. The second CFH pair is also connected to the mirror, so when the MMVCM exerts force on the mirror, the rotational displacement of the hinges results in a change in the mirror's angle.

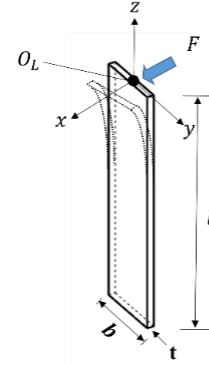


Fig. 5. Deflection of leaf spring

Fig. 5 shows leaf springs that configure the CFH proposed in this study. Four leaf springs configure the CFH as shown in Fig. 4. The springs has a thickness, length, width, and intervening angle. Depending on the allowable rotation angle, the pairs will have an allowable gap with the midbody or base. Each design parameter on Table 1. is related to the performance of the flexible hinge. The appropriate value will be selected by the optimization design.

TABLE I. DESIGN PARAMETERS OF CFH

symbol	Description
l	Length of leaf spring
t	Thickness of leaf spring
b	Width of leaf spring
d	Clearance for motion

The deformation of the leaf springs δX in response to external force F is as follows [9];

$$\delta X = CF \quad (1)$$

$$\delta X = [\delta_x \delta_y \delta_z \theta_x \theta_y \theta_z]^T, F = [F_x F_y F_z M_x M_y M_z] \quad (2)$$

$$C = \begin{bmatrix} c_1 & 0 & 0 & 0 & c_3 & 0 \\ 0 & c_2 & 0 & -c_4 & 0 & 0 \\ 0 & 0 & c_5 & 0 & 0 & 0 \\ 0 & -c_4 & 0 & c_6 & 0 & 0 \\ c_3 & 0 & 0 & 0 & c_7 & 0 \\ 0 & 0 & 0 & 0 & 0 & c_8 \end{bmatrix} = K^{-1} \quad (3)$$

$$c_1 = \frac{l^3}{3EI_{zz}}, c_2 = \frac{l^3}{3EI_{yy}}, c_3 = \frac{l^3}{2EI_{zz}}, c_4 = \frac{l^3}{2EI_{yy}},$$

$$c_5 = \frac{l}{Ebt}, c_6 = \frac{l}{EI_{zz}}, c_7 = \frac{l}{EI_{zz}}, c_8 = \frac{l}{Gkb^3t}$$

δX represents the 6-directional deformation of the leaf spring. C represents the compliance matrix which is the inverse of the stiffness matrix K . G , E and I_{ij} represent the shear modulus, young's modulus, and area moment of inertia ($i, j = x, y, z$), respectively. The twisting factor k is set to 0.3. Using the 6-DOF deformation of the leaf spring with the translation and conversion matrix, the deformation of the first and second CFH pairs is calculated. Rotational and transformation matrix from the coordinate on the base (O) to each the leaf spring (O_L) are as follow;

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi_x & -\sin \phi_x \\ 0 & \sin \phi_x & -\cos \phi_x \end{bmatrix} \quad (4)$$

$$R = R_z R_y R_x \quad (5)$$

$$P = \begin{bmatrix} 0 & -p_z & p_y \\ p_z & 0 & -p_x \\ -p_y & p_x & 0 \end{bmatrix} \quad (6)$$

Eq. (4) and (6) shows the rotational matrix and position matrix. F' is external force on the body's origin (O') and δp is the body's displacement.

$$F' = \begin{bmatrix} I & 0 \\ P & I \end{bmatrix} F \quad (7)$$

$$\delta p = \begin{bmatrix} I & P^T \\ 0 & I \end{bmatrix} \delta X \quad (8)$$

Eq. (7) and (8) lead to a new compliance matrix when considering body's displacement (X).

$$C_{rot} = \begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix} C \begin{bmatrix} R^T & 0 \\ 0 & R^T \end{bmatrix} \quad (9)$$

$$C_{tr} = \begin{bmatrix} I & P^T \\ 0 & I \end{bmatrix} C_{rot} \begin{bmatrix} I & 0 \\ P & I \end{bmatrix} = C' \quad (10)$$

$$X = C' F' \quad (11)$$

From the rotational matrix C_{rot} and transformation matrix C_{tr} , it is possible to determine the amount of displacement of a body that is far from the origin connected by the hinge. C' is the compliance of 1st and 2nd pair CFH's, which is the inverse of the stiffness K' .

C. Definition of the optimal design problem

In this section, we present the problem definition for the optimal design of the 1st and 2nd pair of CFH proposed in this study. The reciprocal of the squared stiffness is the cost function as shown in Eq.(10). The reason for squaring is to deepen the cost function into a gentle convex, making it easier to find the global minimum.

$$\text{Cost Function} = \min \left(\frac{1}{K_{H-CFH}^2} \right) \quad (12)$$

Each displacement of the CFH is inversely proportional to the stiffness and proportional to the driving torque. The stiffness is inversely proportional to the square of the leaf spring's length, proportional to the width, and inversely proportional to the square of the thickness. Before performing the optimal design, the following constraints are given. First, the horizontal and vertical lengths of the CFH should not exceed 13 mm. A pair of CFHs that can be attached to a small FSM should not exceed the mirror area excluding the area occupied by two pairs of MMVCMs. Second, the width of each CFH should not affect the drive range. To ensure the length of the gap while considering the length of the entire mechanism and the two pairs of CFH, the width of the CFH is limited up to 1.5 mm.

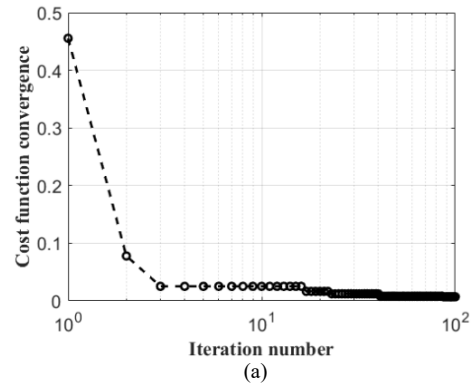
D. Optimization of CFH

Optimization direction was determined by the Broyden-Fletcher-Goldfarb-Shanno (BFGS) method, and the optimization step size was determined by the sequential quadratic programming (SQP) algorithm [10]. The properties of Flexure used for optimization are shown in Table 2.

TABLE II. MATERIAL PROPERTIES OF CFH

Material	Ti-6Al-4V (Grade 5)
Shear modulus (G)	44 GPa
Tensile yield stress	880 MPa
Density	4430 kg/m ³
Young's modulus (E)	113.8 GPa

The optimization process were shown in Fig. 6. The iteration was about 100 times and the resulting cost function was shown in Fig. 6(a).



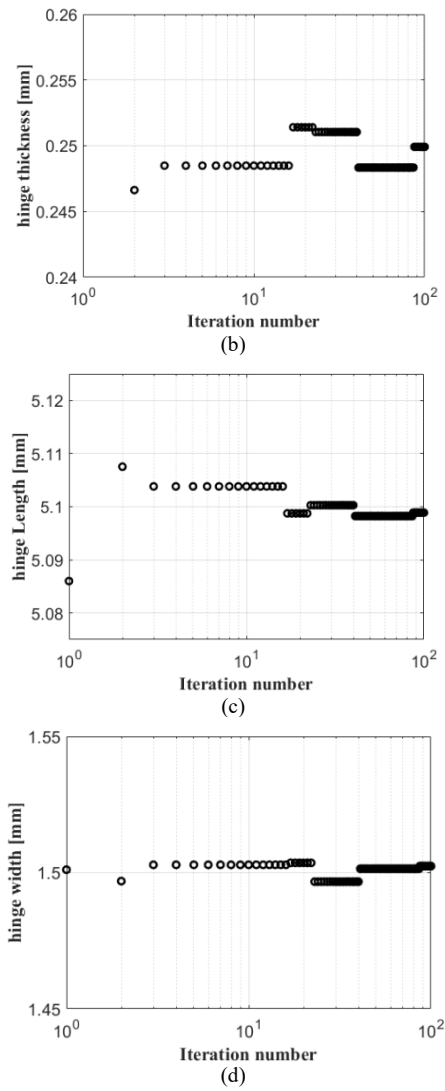


Fig. 6. Optimization process of CFH, (a) cost function, (b) thickness of hinge, (c) Length of hinge, (d) width of hinge

The optimal design result of applying the Global Search algorithm to each pivot is the same as shown in Fig. 6. It shows optimal design results and final design parameters.

TABLE III. OPTIMAL DESIGN RESULTS AND FINAL DESIGN PARAMETERS

Parameters	Optimal design	Final design
Length of leaf spring (l)	5.095 mm	5.1 mm
Thickness of hinge (t)	0.252 mm	0.25 mm
Width of hinge (b)	1.497 mm	1.5 mm

When calculated with the actuator's force and the CFH pair's stiffness, the rotational displacement of the mirror is about 75.8 mrad. When the mirror rotates, however, a parasitic translation of 0.97 mm occurs along x-axis. This is because the mirror rotates at a position 15.7 mm away from the rotation center of the hinge. There must be compensation when performing feedback control.

E. Finite Element Method Analysis

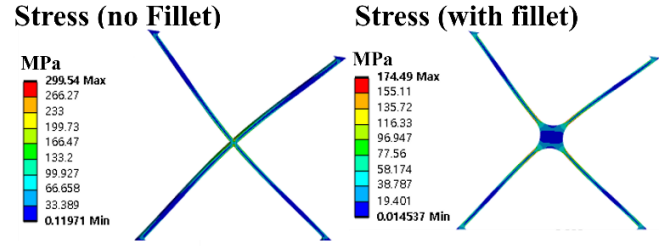


Fig. 7. Stress analysis of a pair of CFH

The rotational motion of CFH was verified by Ansys static analysis. The height of the triangle with the angle between the tilted mirror and the ground is 4.336 mm. the length of the hypotenuse is 64 mm, which is the diameter of the dummy mirror. The calculation result is as follow;

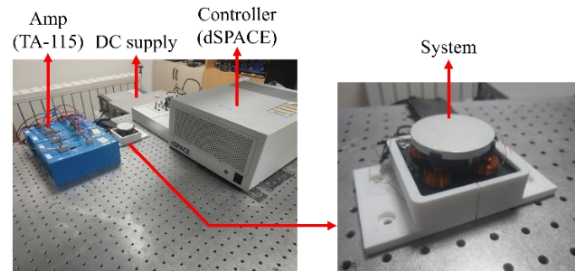
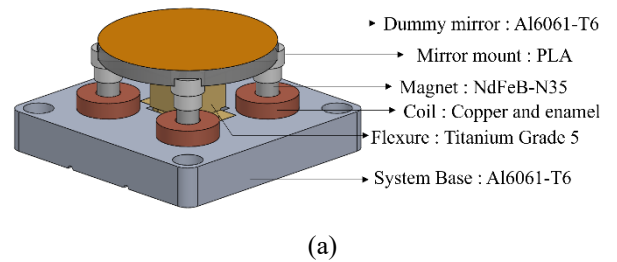
$$\sin^{-1}\left(\frac{4.636 \text{ mm}}{64 \text{ mm}}\right) = 72.5 \text{ mrad} \quad (13)$$

The calculated value and the analytical result show an error of about 4.36%, which is small enough to consider the modeling and design are similar to each other.

Stress concentration occurs at the rotation center (pivot) when it rotates. Because the allowable stress is about 30% of the yield stress, FEM analysis needs to check whether the stress is lower than the allowable stress or not. Then, the stress applied to the hinge is shown in Fig. 7. The maximum stress applied to the CFH is 174.5 MPa, which corresponds to 19.82% of the yield stress. So, the stress is allowable. This relaxed stress is due to the fillets in the edges where each leaf spring is connected. Therefore, the design is safe and reliable.

III. MANUFACTURING RESULTS AND PERFORMANCE VERIFICATION

A. Final design of the Mechanism



(b)

Fig. 8. Fabricated FSM, (a) schematics, (b) experimental setting of system and fabricated FSM

The fabricated FSM mechanism is presented in Fig. 8. The mirror is actuated by two pairs of MMVCM in a pull-push mode. The maximum torque of one pair of MMVM is 89.6 N/mm.

B. Control Strategy

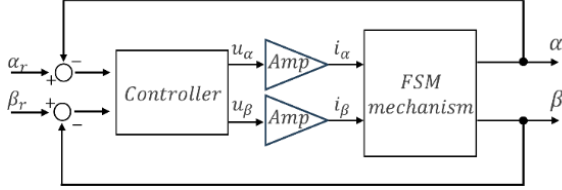


Fig. 9. Control Diagram of FSM

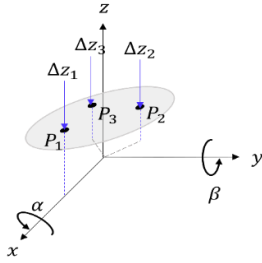


Fig. 10. Schematic diagram of sensor kinematics

As shown in Fig. 9, the control inputs are α_r and β_r , which are the desired angular motion of the mirror about x-axis and y-axis. The errors obtained by subtracting the actual angular motion of the mirror from the inputs are given to the controller. Each the pair of MMVCM generates a torque, and the torque in equilibrium with each the pair of CFHs become the actual α and β . Fig. 10 is a schematic diagram of the kinematics of the sensor. Each of the laser displacement sensors that measure the height of the mirror from their base.

C. performance verification

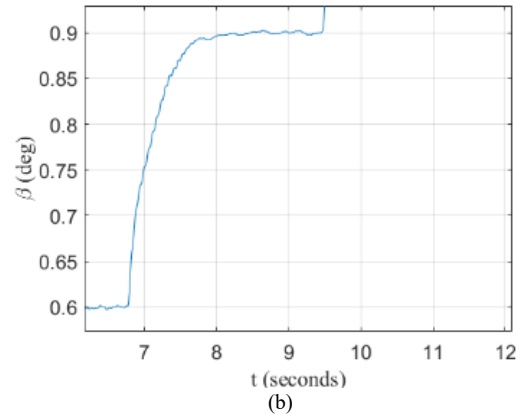
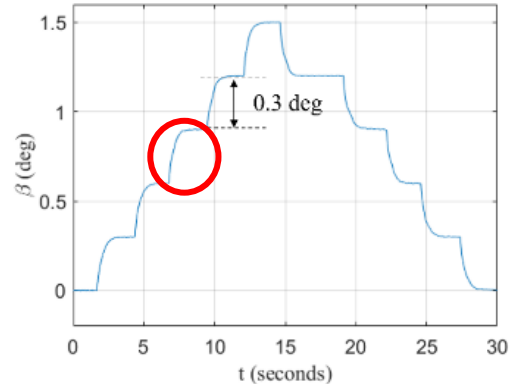
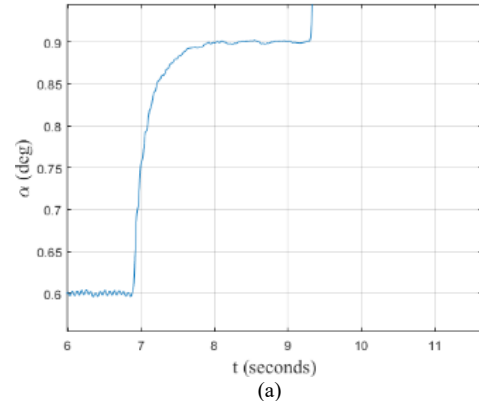
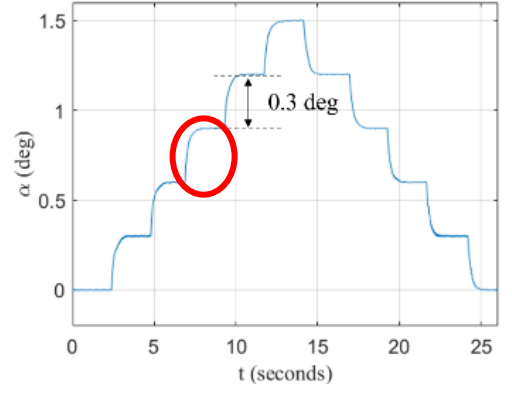


Fig. 11. Results of driving range test; (a) Step response of Alpha, (b) Step response of Beta

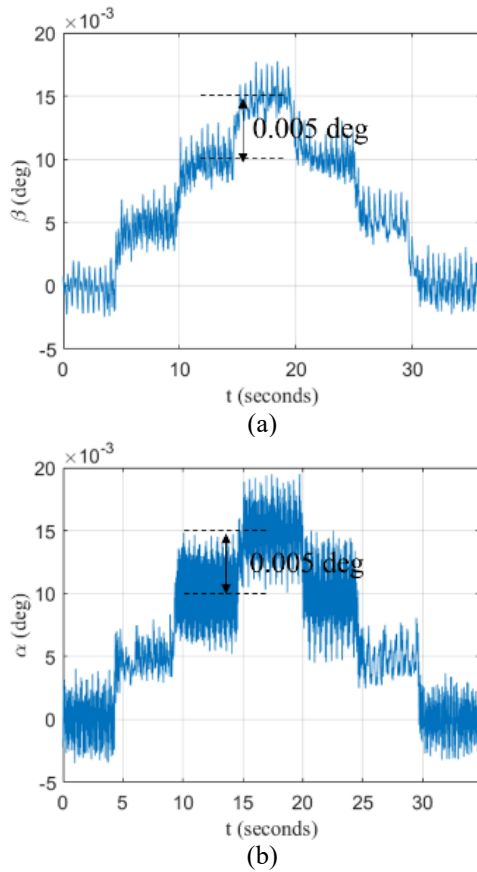


Fig. 12. Results of resolution test; (a) resolution of Alpha, (b) resolution of beta

To test the working range of the FSM mechanism, step response was investigated. At this time the mechanism was feedback-controlled. The step input was in steps of 0.3° up to 1.5° , and the results are the same as in Fig. 11(a) and (b). The results of control for α and β are shown in Table 4 below.

TABLE IV. RESULTS OF CONTROL

Performance variable	α	β
Driving range	1.5°	1.5°
Rise time	0.8203 s	0.5004 s
Overshoot	0 %	0 %
Settling time	1.004 s	1.4 s
Steady state error	0.164 %	0.254 %

To examine the repeatability of the FSM, minimum step response was investigated. The results are the same as Fig. 12(a) and (b). As results of the control, a resolution of 0.005° was achieved. It is expected that the resolution can be further improved if the problem of sensor noise and environmental disturbances are resolved.

IV. CONCLUSION

In this paper, we designed a Fast Steering Mirror(FSM) mechanism for laser beam alignment of Free Space Optical communication. Specially, a cartwheel flexural hinge was applied as a rotational guide. The hinge design is optimized to

maximize its stiffness. We also verified the model and design through Finite Element Method. The performance of FSM mechanism was investigated by feedback control for the fabricated Fast Steering Mirror mechanism. The experimental results showed that the system achieved a rise time of 0.786 s, an overshoot of 0%, a settling time of 2.1 s, and a steady state error of less than 1% within a driving range 1.5° . The resolution was 0.005° .

This confirms the FSM mechanism's applicability to large-area/precision Fast Steering Mirror. It is necessary to compensate for low-frequency sensor noise and random environmental disturbances. The mechanism has the potential to be applied to LIDAR (Light Detection and Ranging), Space telescope, and so on.

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